

STOCKSWISE PROJECT · CS-480 — DATA VISUALIZATION

StocksWise

A beginner-friendly, data-driven guide to investing on the NASDAQ.

EPFL

Process Book Milestone 3

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1. Overview of the project

The primary goal of the StocksWise project is to demystify the stock market for newcomers and to provide a single, coherent learning path that bridges the gap between purely conceptual finance content and the dry, numeric tools used by professionals. Our objective is to offer accurate, real-data visualizations paired with progressive explanations that make the NASDAQ approachable for someone who has never bought a share.

1.1. OUR GOAL

The overarching theme of this project is the intersection of retail investing and data literacy. The modern stock market is overwhelmingly noisy: financial media is fragmented between sensationalist news, opaque jargon, and broker dashboards that throw a hundred indicators at users without explaining any of them. Beginners trying to invest their first 1,000 face a paradox the more they read, the less confident they feel. Our objective is to quantify, visualize, and explain the core mechanics of the market so that an interested non-expert can build an accurate mental model in a single afternoon. By providing detailed yet readable visualizations of real NASDAQ data, we aim to encourage informed decision-making and to push back against the gambling-like framing that dominates much of online stock content.

This project aims to offer meaningful insights not only for beginners but also for intermediate users who want to deepen their understanding of risk, diversification, and historical market behavior. By combining storytelling with real data, we engage a broad spectrum of users: students, casual investors, finance-curious readers in a clearer dialogue about what investing actually means.

1.2. OUR CONTRIBUTION

The project is developed as a single-page educational website complemented by a dedicated simulator page. It features four learning modules *Stocks*, *ETFs*, *Risks*, and *Simulation* each built around a set of carefully chosen, interactive D3.js visualizations. A persistent **Beginner / Advanced toggle** reframes the entire site at any moment: the same data, the same charts, but explanations that shift from intuition-first to formula-first. This dual-mode design is, to our knowledge, uncommon in finance learning sites, and it is the central pedagogical contribution of the project.

2. From concept...

2.1. PREMISES OF THE PROJECT

Our project began with a simple observation: among our peers at EPFL, almost everyone we asked had thought about investing, but very few had actually started. The reasons given were always the same *“I don’t really understand what a stock is”, “I’m scared of losing everything”, “I don’t know where to begin”*. These are not technical barriers, they are literacy barriers. We set out to build a tool that takes a user from *“I have 1,000 in a savings account”* to *“I understand what a portfolio is and why I’d build one.”*

The initial phase of our project involved deciding the angle. After brainstorming, we converged on the idea that a *guided narrative*, organized around a hypothetical 1,000 to invest, would be far more effective than a reference-style site. Each module would answer a specific question a beginner asks themselves *What am I buying? How do I avoid putting all my eggs in one basket? What can go wrong? What would have happened if...?*

2.2. SELECTING THE CORRECT DATA

When selecting our dataset, we asked ourselves: *“What is the story we want to tell?”* We surveyed existing finance-education sites and noticed two failure modes: either they show real

data with no explanations (Yahoo Finance, TradingView), or they explain concepts without ever showing real data (most beginner blogs). We wanted both, on the same screen, at the same time.

Our primary dataset is built on 45 years of historical NASDAQ daily price data covering more than 6,800 tickers. We focus the user-facing visualizations on a curated subset of the most recognizable NASDAQ names AAPL, MSFT, NVDA, AMZN, TSLA, GOOG, META, AMD plus the QQQ ETF as a proxy for the index itself. This choice was deliberate because beginners learn faster when the companies on screen are ones they already know. We also include sector classifications (drawn from public listings) to power the diversification module.

To support the historical analysis and simulation features, we identified five major crisis windows *1987 Black Monday*, *the 2000 Dot-com Bubble*, *the 2008 Financial Crisis*, *the 2020 COVID Crash*, and *the 2022 Rate-Hike Selloff* and curated annotation data for each. These are the moments that shaped the modern market, and being able to “rewind” to them is, we believe, one of the most powerful pedagogical tools the site offers.

The story accompanying the visualizations guides users from the atomic unit (one stock, one chart) up to portfolio-level thinking (correlation, efficient frontier), then to risk awareness (volatility, drawdown, Sharpe), and finally to historical perspective (the simulator). By interacting with each module, users build the next one’s intuition without realizing it.

3. ...to beginner mode, learning the basics of investing

The Beginner Mode was designed as the educational entry point of StocksWise. Its objective is to introduce fundamental investing concepts through simple and intuitive visualizations, without overwhelming users with technical jargon or complex financial indicators.

The mode is structured around three beginner-oriented sections: *Stocks*, *ETFs* and *Risks*. Each section focuses on answering a practical question that a first-time investor might ask, such as “*What is a stock?*”, “*Why should I diversify?*”, or “*What happens if the market falls?*”

The **Stocks** section introduces users to the basic mechanics of the stock market. Instead of immediately displaying complex charts, the page first explains essential concepts such as what a stock represents, how stock markets function, and how prices are determined through supply and demand.

The interface was intentionally designed around short educational cards and simple terminology to make finance feel approachable for users with no prior investing experience. Key financial terms such as ticker symbols, market capitalization, volume, and adjusted close prices are progressively introduced in a lightweight format.

The **ETFs** section focuses on diversification and portfolio construction. Through intuitive portfolio comparisons and sector visualizations, users progressively understand why diversified investments behave differently from concentrated portfolios. The goal here is not mathematical precision, but visual intuition.

The **Risks** section introduces beginner users to the concept of volatility and market uncertainty. Instead of presenting advanced metrics immediately, the visualizations focus on helping users understand that investing also involves losses, fluctuations, and emotional reactions during crises.

Finally, we added also a **Simulation** section, apart from the modules, that allows users to create a simple portfolio and observe how their investment evolves over time. Users can select assets, allocate weights, and visualize portfolio performance directly on historical NASDAQ data. This simplified simulation experience acts as a first introduction to long-term investing behavior.

To reinforce learning, quizzes were integrated throughout the Beginner Mode. Rather than testing memorization, these quizzes encourage users to interpret the visualizations they just explored, making the learning process more interactive and engaging.

4. ...to advanced mode, deeper financial analysis

While the Beginner Mode focuses on accessibility and financial literacy, the Advanced Mode transforms StocksWise into a more analytical and data-driven platform. This mode was designed for users who want to explore financial markets through professional-style visualizations and interactive analytical tools built on real NASDAQ data.

The Advanced Mode is divided into four major sections: *Technical Analysis*, *Portfolio Construction*, *Risk Metrics Deep Dive*, and *Market History & Psychology*. Together, these sections progressively move the user from individual stock analysis to broader portfolio and market-level understanding.

The **Technical Analysis** section introduces users to the visual tools commonly used to analyze stock-price behavior. The main visualization is an interactive candlestick chart enhanced with multiple technical indicators such as moving averages, RSI, and support/resistance levels.

Rather than simply displaying static indicators, the interface allows users to dynamically activate overlays and explore how different indicators react to real historical market conditions. The objective of this section is to transform technical analysis from abstract finance terminology into an interactive visual exploration experience.

The **Portfolio Construction** section focuses on diversification and asset allocation. Through correlation heatmaps, sector distributions, and efficient-frontier visualizations, users can understand how different assets interact within a portfolio and how diversification affects overall portfolio behavior.

One of the key visualizations of this section is the efficient-frontier plot, which illustrates the trade-off between return and risk across different portfolio allocations. Combined with the correlation heatmap, the section demonstrates that portfolios containing many technology stocks may still remain highly correlated internally.



Figure 1: Portfolio-construction visualizations in Advanced Mode.

The **Risk Metrics Deep Dive** section introduces users to more advanced portfolio-analysis metrics. Instead of focusing solely on returns, this section emphasizes the importance of volatility,

drawdowns, and risk-adjusted performance.

The visualizations include rolling volatility charts, maximum drawdown analysis, Sharpe ratio indicators, and strategy-comparison equity curves. By interacting with these visualizations, users can directly observe how different strategies behave during unstable market periods and why risk management is a central component of investing.

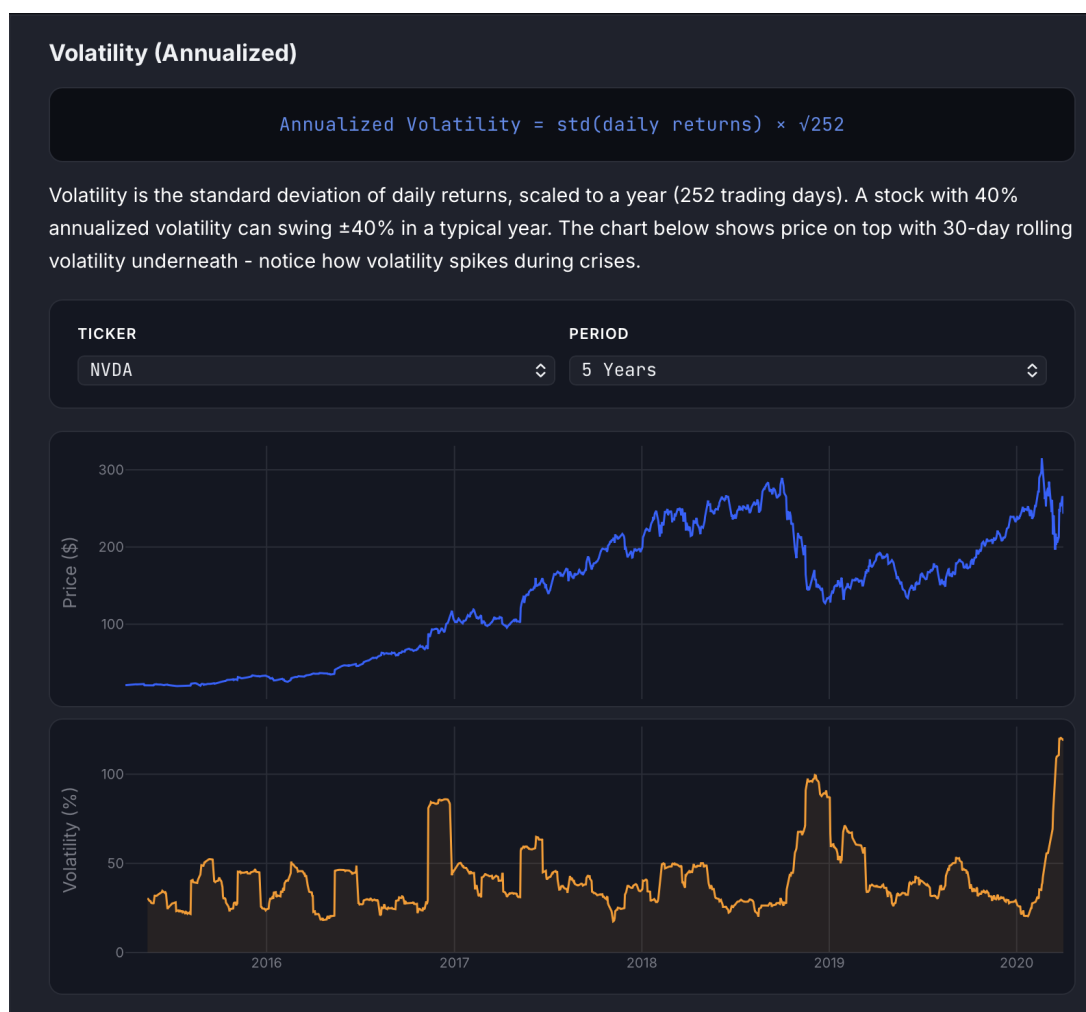


Figure 2: Risk-analysis visualizations based on historical NASDAQ data.

The **Market History & Psychology** section focuses on understanding how financial markets react during major historical events and periods of uncertainty. Through interactive simulations and historical replay tools, users can explore how portfolios behave during major market crises such as the Dot-com Bubble, the 2008 Financial Crisis, or the COVID crash.

This section combines portfolio simulation, historical timelines, crisis overlays, and strategy backtesting into a single interactive experience. The objective is not only to analyze financial performance, but also to help users understand the psychological dimension of investing during periods of extreme uncertainty and volatility.

Watching portfolios evolve through historical crises allows users to experience the asymmetry of market crashes and recoveries in a much more intuitive way than through theoretical explanations alone.

A key challenge throughout the Advanced Mode was balancing analytical depth with readability. Financial indicators and simulations can quickly become visually overwhelming when multiple datasets and overlays are displayed simultaneously. To address this, every visualization was designed around progressive interaction and optional layers, allowing users to explore complex financial information while maintaining a clean and understandable interface. Unlike the simplified simulation integrated into the Beginner Mode, the Advanced simulation tools focus on historical crisis analysis, strategy backtesting, and portfolio stress testing.



Figure 3: Portfolio-construction visualizations in Advanced Mode.

5. Breakdown

5.1. GUSTAVE

Gustave mainly focused on the coding aspect of the project. He handled the integration between the website and the data, ensuring that all components worked smoothly and in synchronization.

5.2. MAXIME

Maxime was primarily responsible for designing the main plots and developing the website's user interface and overall user experience, including the simulator.

5.3. YOUSSEF

Youssef mainly handled the technical content of the website, from designing the overall data narrative to finding innovative and visualization-driven ways to present it.

5.4. THE FINAL WORD

Working on StocksWise has been a great opportunity for us to combine our skills in data engineering, web development, financial modeling, and design. We are proud of the result a coherent, dual-mode learning experience that we believe meaningfully lowers the barrier to entry for someone curious about the stock market. Most importantly, the project gave us the chance to discuss what makes a visualization *educational* rather than just *decorative*, and to put those discussions into practice. Enjoy and remember that this is not financial advice.